ZIP 29492, Daniel Island and Cainhoy, Charleston

An El Niño winter here means nor'easters, heavy rain at high tide, and freezes that close bridges. The deaths come in cars driven into water, and in people cut off from power, dialysis, or warmth. The first move is to close the flooded roads before anyone drives into them.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
A Berkeley County, SC region (inferred; not resolved)	None published for the US or South Carolina	Sep 2026 – Feb 2027 (9-month study: Jun – Feb)	Flood (rain and tide together), then cold and power loss

14/14 recommendations verified. 10 corrected, 0 unverified, 1 contraindicated. Two items were contraindicated as first drafted: a road-closure trigger that would have fired on ordinary spring tides (corrected, action 1), and a request to SC DSS for a disaster food plan the law already requires (moved to Do not do this). Exposure figure: Climate Impact Lab, September 2026, pre-peer-review, heat only, with no figure published for the United States. Heat is not this place's winter hazard, so the historical record carries the risk here. The funding target fell from the first draft's $437,664 to $109,400 in verification, because the unit cost behind most of it was a national cooling-bill average, not a heating figure.

EXECUTIVE SUMMARY — AT A GLANCE



32.88°N 79.91°W, United States of America · Natural Earth

- **The risk.** The El Niño nor'easter of 17 Dec 2023 drove the harbour to 9.86 ft, the highest tide ever recorded without a hurricane, and a 72-year-old woman drowned in her car in Mount Pleasant.
- **For you.** Never drive into a flooded road, fit a $20–40 CO alarm by 30 November, and register your cell phone with Berkeley County's alerts.*
- **The money.** $23,960 floor, $109,400 target, $252,280 ceiling; the cheapest lives saved are paid overnight staff at the Aldersgate warming shelter and CO alarms.
- **Do not.** Do not rely on "Turn Around Don't Drown" signs alone: no study shows they cut flood-driving deaths, and only a closed road does.

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* Suggested formulation: check with a health worker before sharing.

WHERE THIS IS

SCOPE

This briefing covers ZIP 29492: Daniel Island and the Cainhoy peninsula along Clements Ferry Road, all inside the City of Charleston and Berkeley County. Beeple Studios, at 2325 Charleston Regional Parkway, sits on the Cainhoy side, north of I-526 and about 6 km from the island itself. Two requests were made, one for the studio and one for Charleston; this briefing answers both. It sets the ZIP inside the wider city, because the people most at risk in a Charleston El Niño winter mostly live elsewhere, in North Charleston (29405) and on the peninsula (29403). A separate briefing for either of those can be built on request.

The population figures below are for Berkeley and Charleston counties or for whole ZIP codes, as published. None is a Daniel Island figure unless it says so.

FOR YOU, AS A HOUSEHOLD

9 STEPS

For a household in 29492, an El Niño winter means more nor'easters, heavy rain landing on high tides, and the odd hard freeze that closes the bridges. People here die in cars driven into flooded roads, and at home when the power, the heat, or the dialysis appointment is cut off. The ranked actions below are written for the city, the counties, and their partners. These nine are the ones you can take yourself, at your own cost, grouped by what they protect. Added 25 October 2026 at a reader's request.

1 · Floods: never drive into water on a road, and check the city's flood maps before any evening trip when a flood warning meets a high tide

PARTIALLY CONFIRMED FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
Now, all winter	$0	Everyone in the car

Vehicle drownings are how floods kill here: on 17 December 2023 a 72-year-old woman drowned in her car on Omni Boulevard in Mount Pleasant, during a downpour on a record tide. The National Weather Service puts it plainly: 12 inches of rushing water can carry away most cars, and 2 feet can carry away SUVs and trucks. Before driving when a coastal-flood or flash-flood warning is out, look at TIDEeye (gis.charleston-sc.gov/tideeye) and the regional road-closure map (gis.charleston-sc.gov/road-closures-regional/map). Never drive round a barricade; Charleston police said in October 2026 that they will ticket drivers who do. If you cannot see the road surface, turn around.*

* Suggested formulation: check with a health worker before sharing.

2 · Carbon monoxide: fit a battery alarm outside each sleeping area and on every level by 30 November, and never run a generator or grill indoors, in the garage, or by a window

PARTIALLY CONFIRMED · FROM ACTION 5

BY WHEN	COST TO YOU	PROTECTS
30 Nov 2026	$20–40 per alarm	Everyone asleep in the house

After a 2002 ice storm in Mecklenburg County, North Carolina, 88% of people poisoned by carbon monoxide, and 25 of the 26 severe cases, were in homes without a working alarm. Grills were a source as well as generators. Cases peak in the first three nights of an outage, mostly overnight. Keep a generator outdoors, at least 20 feet from doors, windows, and vents (CDC). The Red Cross and Charleston Fire install smoke alarms, not CO alarms, so buy your own.*

3 · Power: check which utility you are on, make sure its outage texts reach your phone, and if anyone at home depends on medical equipment, file a medical certificate before 1 December

PARTIALLY CONFIRMED · FROM ACTION 3

BY WHEN	COST TO YOU	PROTECTS
1 Dec 2026	$0; about $30–150 for lights and a power bank (estimate)	Oxygen, medical devices, phones

Daniel Island is mainly Dominion Energy South Carolina; parts of Cainhoy may be Berkeley Electric Cooperative, so check your address on its service map. Dominion customers with a smart meter and a mobile number on file get outage texts automatically from short code 64312; check them under Profiles and Settings, then Notification Preferences. Berkeley Electric's outage line is 1-855-938-3615. A medical certificate, signed by a licensed health care provider and filed with the utility, should stop disconnection for non-payment from 1 December to 31 March. The bill is still owed. Keep flashlights rather than candles, a charged power bank per phone, and a backup battery for any powered medical device.*

4 · Staying warm and keeping the pipes whole: know where the warming sites are, heat one room, and drip your faucets on freezing nights

CONFIRMED · FROM ACTION 4

* Suggested formulation: check with a health worker before sharing.

BY WHEN	COST TO YOU	PROTECTS
Before the first freeze	$0–30 (pipe insulation, blankets)	Older family members, and the house

None of last winter's warming sites is in 29492. The nearest are Aldersgate in North Charleston (Holy City Missions) and Hibben UMC in Mount Pleasant (Sundays only); check what is open before you need it. If the power is off and the house is cold, gather in one small room with blankets over the doorways, or leave. Charleston Water System's freeze advice is to let faucets drip about once a second, insulate exposed pipes and backflow preventers, and disconnect hoses. In the January 2018 freeze there were more than 70 main breaks, and demand jumped from about 60 to 94 million gallons a day. CWS asked customers to conserve, and credits that with keeping pressure up. So once your pipes are protected, use as little water as you can.*

5 · Food: keep three to five days of food that needs no cooking and no fridge, for the days when the bridges are closed

PARTIALLY CONFIRMED · FROM ACTION 8

BY WHEN	COST TO YOU	PROTECTS
15 Dec 2026	$40–60 for thermometers and a cooler; about $50–75 a person for food (estimates)	Your household's meals

Food reaches 29492 by truck over I-526, I-26, and the river bridges, so the weak point is a closed road, not empty shelves; in January 2018 the I-526 ramps and the Wando bridge on Highway 41 closed. USDA's rule for outages: a closed fridge keeps food safe about 4 hours, a full freezer 48, a half-full one 24. Throw out perishables above 40°F for more than 2 hours. A few days of supplies is enough; once the roads reopen, the shops work.

6 · Water: store at least a gallon per person per day, for several days, and keep it if a boil-water notice comes

CONFIRMED · ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
30 Nov 2026	About $20–40 for containers (estimate)	Drinking water

Ready.gov advises at least one gallon per person per day, for several days; count pets too. In the December 2022 freeze, with thousands of leaks and use near 100 million gallons a day, Charleston Water System warned that a boil-water notice might follow. Stored water

* Suggested formulation: check with a health worker before sharing.

covers you if one does.*

7 · Communications: register your cell phone and email with Berkeley County's alert system, keep a NOAA weather radio, and agree an out-of-area contact

CONFIRMED FROM ACTION 6

BY WHEN	COST TO YOU	PROTECTS
15 Nov 2026	$0; about $25–60 for a radio (estimate)	Contact when phones and internet fail

All of 29492 is in Berkeley County, so the system is the Berkeley County Emergency Notification System (berkeleycounty.onthealert.com), not Charleston County's. Landlines are already in it; add your cell phone and email yourself. The NOAA Weather Radio station for Charleston is KHB-29 on 162.550 MHz, with WXJ-23 (Green Pond, 162.450) as backup. Agree on one relative out of state whom everyone texts, because long-distance texts often get through when local calls fail. On Daniel Island, the property owners' association takes storm problems at reportaproblem@dicommunity.org. Know the two or three neighbours who live alone, and check on them.

8 · Health: flu and COVID shots by 15 November, the one-time RSV dose if you qualify, a week of prescriptions on hand, and an emergency plan with your dialysis clinic if you need one

PARTIALLY CONFIRMED FROM ACTION 2

BY WHEN	COST TO YOU	PROTECTS
15 Nov 2026	Usually $0 if insured	Lungs, hearts, kidneys

Major medical societies (AAP, AAFP, ACOG, and IDSA, September 2026) recommend a flu shot every year for everyone 6 months and older, and a COVID shot every year for all adults. South Carolina pharmacists may give COVID vaccine without a prescription. RSV vaccine is one lifetime dose for everyone 75 and over, and for ages 50 to 74 at increased risk; being on dialysis counts. If anyone in the household is on dialysis, ask the clinic now for its emergency plan, early dialysis before a forecast storm or freeze, and the 3-day emergency diet. After Hurricane Sandy, patients dialysed early had lower 30-day mortality (adjusted odds ratio 0.72).*

* Suggested formulation: check with a health worker before sharing.

9 · Money: buy or renew flood cover before 11 December, renters included, and book a heating-help appointment if you qualify

PARTIALLY CONFIRMED | FROM ACTION 7

BY WHEN	COST TO YOU	PROTECTS
Flood cover before 11 Dec 2026	Premium varies; $0 to apply for help	The house, its contents, and the heating bill

A new flood policy takes 30 days to start, and the federal flood programme's authority runs only to 11 December 2026 under the current stopgap budget. If it lapses, no new policies can be sold, though existing ones keep running. In the October 2015 floods only 10% of Berkeley County homes had flood cover. Renters can buy contents-only cover. Standard homeowner and renter policies exclude flood. For heating bills, LIHEAP in Berkeley County runs through Palmetto Community Action Partnership: book at palmettocap.itfrontdesk.com or call 843-724-6760. Help depends on funds, so apply early.

What to ask of the people in charge

- Ask **City of Charleston Emergency Management** whether the low points on Clements Ferry Road and the I-526 ramps are on the barricade list this winter (action 1).
- Ask the **Daniel Island Property Owners Association** to add a check on older and isolated residents to its storm plan, alongside the property checks (action 6).
- Ask the **City of Charleston Fire Department** whether CO alarms for neighbours who cannot afford them are funded this winter (action 5).

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached

* Suggested formulation: check with a health worker before sharing.

independently is the stronger bet.

1. By 30 November, extend Charleston's flood-closure system to the 29492 approaches, and close roads when a flash-flood warning meets a high tide

PARTIALLY CONFIRMED · EMERGENCY MANAGER · SKEPTIC · PUBLIC HEALTH · HISTORIAN · ADAPTATION INVENTORY: COULD NOT BE CONSULTED

OWNER	BY WHEN	COST	PREVENTS
City of Charleston Emergency Management, with Police and SCDOT	30 Nov 2026	Existing staff and barricades; no new figure	Drownings in vehicles, the main way people die in Lowcountry floods

Charleston already closes flooded streets with barricades, shows closures on its TideEye map, and tickets drivers who go round them. It closed 21 streets in one recent event. The gaps are where the barricades go and what sets them off. Add the low points on Clements Ferry Road and the I-526 ramps, agree the Mount Pleasant segments with the town and SCDOT (Omni Boulevard, where Nancy Morrow, 72, drowned in her car on 17 Dec 2023), and publish dry detours. Keep the tide trigger at the city's 7.5 ft (moderate) level on the Charleston Harbor gauge, NOAA 8665530. Add a second trigger: a National Weather Service flash-flood warning during a high tide. The local deaths and the 78 rescues on 17 Dec 2023 came in a downpour, not from the tide alone. Then assume the tide can come in about 1.5 ft above the forecast, as it did that day (forecast 8.2 to 8.3 ft, actual 9.86 ft).

Transferability: the death pattern is local. Vehicle drownings were 18 of 43 deaths in Hurricane Matthew (2016), but those were in North Carolina, so that figure describes the hazard, not Charleston. Corrected in verification: the first draft's trigger ("within 1 ft of 7.0 ft") equals about 6 ft, roughly an ordinary high tide, and would have closed roads on every spring tide.

2. By 30 November, pre-book early dialysis and transport for the ~734 facility dialysis patients in Charleston and Berkeley counties, to run 48 hours before a forecast storm or freeze

CONFIRMED · PUBLIC HEALTH · ADAPTATION INVENTORY: COULD NOT BE CONSULTED

OWNER	BY WHEN	COST	PREVENTS
IPRO ESRD Network 6, with DaVita, Fresenius, and the two county EMDs	30 Nov 2026	Within existing dialysis and transport budgets	Deaths from missed dialysis when bridges and roads close

* Suggested formulation: check with a health worker before sharing.

HHS emPOWER counts 474 Medicare beneficiaries on facility dialysis in Charleston County and 260 in Berkeley (snapshot of 7 Oct 2026). After Hurricane Sandy, patients given dialysis early had lower 30-day mortality: adjusted odds ratio 0.72 (95% CI 0.52 to 0.997), in 13,836 patients in New York and New Jersey (Lurie et al., *Am J Kidney Dis* 2015). Missed sessions were driven by transport and unstable housing. For 29492, the danger is the bridges: in the January 2018 ice storm the I-526 ramps and the Wando bridge on Highway 41 closed.

Transferability: good. It is the same hazard class (a coastal storm), and early dialysis is already standard hurricane practice in the Southeast; this extends it to winter. The study is observational, and the unadjusted result was not significant. Network 6 can coordinate but cannot make a clinic accept a patient.

3. By 1 December, help power-dependent residents file medical certificates with their utility, and plan outreach from emPOWER's counts

PARTIALLY CONFIRMED | PUBLIC HEALTH · ECONOMIST | ADAPTATION INVENTORY: COULD NOT BE CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Berkeley County EMD, with Charleston County EMD and Dominion Energy SC	**1 Dec 2026**	**Staff and volunteer time**	**Deaths when oxygen, ventilators, or heat fail in an outage or shutoff**

emPOWER counts 2,648 Medicare beneficiaries using powered medical equipment in Charleston County and 1,523 in Berkeley. In 29492 it is 117 of 4,378 (2.7%); in North Charleston (29405) it is 200 of 4,573 (4.4%), the highest share of the three ZIPs checked. Two steps work before winter. First, a doctor-signed medical certificate filed with Dominion Energy SC, the main utility on Daniel Island, blocks disconnection for non-payment from 1 December to 31 March, until it expires. Second, regulated utilities may not disconnect when the average forecast temperature is 32°F or below for 48 hours. Residents on Berkeley Electric Cooperative or Santee Cooper, which may include parts of Cainhoy, should ask their utility, because those rules were not confirmed for them. LIHEAP heating help runs through Palmetto Community Action Partnership (Berkeley office 843-761-6935).

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: a county cannot get names from emPOWER. Named lists go only to the state health authority (SC Department of Public Health), through HHS, during a declared incident. The counts are for planning. The harm evidence, about 90 excess deaths against 6 officially counted in the 2003 New York blackout (Anderson and Bell, *Epidemiology* 2012), comes from a summer blackout in a different city.

4. By 1 December, agree one winter shelter plan that pays for overnight staff and runs evening pickups from the Cainhoy and Daniel Island side

PARTIALLY CONFIRMED | EMERGENCY MANAGER · FRONTLINE · PHILANTHROPIST · PUBLIC HEALTH | ADAPTATION INVENTORY: COULD NOT BE CONSULTED

OWNER	BY WHEN	COST	PREVENTS
City of Charleston, with Holy City Missions, One80 Place, the Salvation Army, and Berkeley County	1 Dec 2026	$21,600 for paid overnight staff (estimate)	Hypothermia deaths among unsheltered people

A 59-year-old Charleston County man, reportedly homeless, died in Winter Storm Fern in late January 2026. Officials believed it was hypothermia, and the coroner called the death storm-related. Last winter's warming sites were Aldersgate in North Charleston (Holy City Missions), Hibben UMC in Mount Pleasant (Sundays only), Seacoast in Summerville, the Salvation Army in West Ashley (50 beds), and Goose Creek UMC for Berkeley County. None is in 29492. Aldersgate, which holds 65 to 75, cannot open without two overnight volunteers, and it struggles to find them. One plan should set a single opening temperature (Holy City Missions used 40°F in 2025 reporting), pay overnight staff so the doors open every cold night, and add a pickup on the Wando side. CARTA and Tri-County Link already run some rides.

* Suggested formulation: check with a health worker before sharing.

Transferability: the transport barrier is well documented: 31% of non-users named it in Maricopa County (CDC MMWR 2025). That was a heat study in Arizona, but the barrier is the same. The 97 unsheltered people in Charleston County is a January 2024 count, from press reports.

5. Before the first outage, have fire departments pair every generator message with a battery CO alarm and "never indoors, in a garage, or near a window"

PARTIALLY CONFIRMED | SKEPTIC · PUBLIC HEALTH | ADAPTATION INVENTORY: COULD NOT BE CONSULTED

OWNER	BY WHEN	COST	PREVENTS
City of Charleston Fire Department, with Berkeley County fire services	1 Dec 2026, then the first 3 nights of any outage	$25,000 for 1,000 alarms at ~$25 (estimate)	Carbon monoxide deaths after the storm has passed

After the 2002 ice storm in Mecklenburg County, NC, 109 of 124 symptomatic CO poisonings (88%) and 25 of 26 severe ones were in homes without a working alarm. Grills were a source as well as generators (CDC MMWR 2004). After the 2004 Florida hurricanes, a portable generator was the source in all five fatal incidents; cases peaked within three days, mostly overnight (CDC MMWR 2005). A Helene death in Chesnee, SC, in 2024 followed the same pattern.*

Corrected in verification: the Red Cross Sound the Alarm programme and Charleston Fire's own programme install smoke alarms only. CO alarms need money of their own; Dorchester County's CO installs are a local model. Transferability: strong. A winter ice-storm outage is the Mecklenburg case.

6. By 30 November, add a people check to the Daniel Island storm plan, and use Beeple Studios for a sign-up night and a weather rule for its own events

PARTIALLY CONFIRMED | FRONTLINE · EMERGENCY MANAGER | ADAPTATION INVENTORY: COULD NOT BE CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Daniel Island Property Owners Association, with Beeple Studios	30 Nov 2026	Volunteer time; ~$0	Isolated older residents going unchecked; crowds driving onto flooded roads at night

* Suggested formulation: check with a health worker before sharing.

The property owners' association starts its storm response about three days out and runs a problem inbox (reportaproblem@dicommunity.org, over 100 emails in Dorian). Its published plan checks property, not people. At the same three-day trigger, volunteers can call a list of older residents who sign up through the newsletter. Beeple Studios, which has held free public events through 2026, can host one evening before December. It would enrol people in the Berkeley County Emergency Notification System (berkeleycounty.onthealert.com; all of 29492 is in Berkeley County), start a phone tree, and hand out a printed card of warming sites, intake times, and pickup points. The studio should also adopt its own rule: cancel or hold an event when a coastal-flood or flash-flood warning overlaps arrival or departure. Across the Wando, the Phillips Community Association (president Richard Habersham) is the contact for nights when the bridges close.

Corrected in verification: the alert system is Berkeley County's, not Charleston County's. The association's plan details date from 2019, and no newer version was found.

7. By 30 November, push flood cover to renters and homes outside the flood zone, while a new policy still starts in time

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: COULD NOT BE CONSULTED

OWNER	BY WHEN	COST	PREVENTS
City of Charleston floodplain management, with the property owners' association	**30 Nov 2026**	**Staff time; premiums paid by households**	**Uninsured flood losses (financial harm, not deaths)**

A new National Flood Insurance Program policy takes effect 30 days after purchase, so cover bought by 30 November is in force from 30 December. In the October 2015 floods only 10% of Berkeley County homes had flood cover, against 93% inside the mapped flood zone (Congressional Research Service). Daniel Island is a mix of high-risk (AE) and moderate-risk (X) zones, with AE along the tidal edges. Renters can buy contents-only cover.

* Suggested formulation: check with a health worker before sharing.

This saves money, not lives, so it ranks below the life-safety actions. Not checked: whether the city's existing flood-insurance outreach already runs this campaign. Parametric cover is not available this winter: South Carolina's August 2026 request for information was market research for a possible state-held policy, not a household product.

8. By 15 December, extend Lowcountry Food Bank's hurricane pre-positioning to winter, stocking partner pantries for 3 to 5 days of closed roads

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: COULD NOT BE CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Lowcountry Food Bank	15 Dec 2026	Not quoted; ask the food bank	Empty pantries in the days after a storm closes bridges

The food bank already stocks partner pantries from April for hurricane season, sized to each agency's clients plus projected disaster demand, and it is a state emergency-response agency with over 230 partners. No winter version was found. Food reaches 29492 by truck over I-526, I-26, and the river bridges, so the weak points are closed roads and freezers that fail in long outages. The bigger pressure on household food this winter is not El Niño. SNAP households fell from July 2025 to July 2026, to 7,458 in Berkeley and 11,685 in Charleston (SC DSS), after federal rule changes.

Corrected in verification: food parcels make sense only for the days when roads are physically cut. After that, SNAP and cash do more, because shops keep working. The El Niño hit to Florida winter vegetables, if it comes, would peak around January to March 2027, after the federal 2026 forecast window. The "9 meals for every 1" line is Feeding America's national SNAP ratio, not a local figure.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	City of Charleston	Berkeley Co. EMD	Charleston Co. EMD	SCDOT	Network 6 & dialysis providers	Shelter operators	Dominion Energy SC	DIPOA	Beeple Studios	Lowcountry Food Bank
1. Flood road closures	**A/R**	**C**	**C**	**R**	**–**	**–**	**–**	**I**	**I**	**–**
2. Early dialysis	**I**	**R**	**R**	**–**	**A/R**	**–**	**–**	**–**	**–**	**–**
3. Power-dependent residents	**C**	**A/R**	**R**	**–**	**–**	**–**	**R**	**R**	**–**	**–**
4. Winter shelter plan	**A**	**R**	**C**	**–**	**–**	**R**	**–**	**–**	**I**	**–**
5. CO alarms	**A/R**	**R**	**–**	**–**	**–**	**–**	**I**	**I**	**–**	**–**
6. People check and sign-up night	**I**	**C**	**–**	**–**	**–**	**–**	**–**	**A/R**	**R**	**–**
7. Flood cover drive	**A/R**	**C**	**–**	**–**	**–**	**–**	**–**	**R**	**I**	**–**
8. Winter food pre-positioning	**C**	**C**	**C**	**–**	**–**	**–**	**–**	**–**	**–**	**A/R**

What the chart shows: the City of Charleston is accountable for four of the eight rows, and Berkeley County Emergency Management is involved in every row but one. Rows 1, 2, 6, and 7 need a decision and staff time, not new money.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR
$23,960

TARGET
$109,400

CEILING
$252,280

LARGEST UNCERTAINTY
Shelter wages and the number of cold nights; no published unit cost exists for either

Floor: 30 nights × 24 staff-hours × $18 ($12,960) plus 500 alarms × $22 ($11,000). Target: 45 nights × 24 × $20 ($21,600), 1,000 alarms × $25 ($25,000), one temporary intake worker for 16 weeks at 40 hours × $20 ($12,800), and $50,000 to a disaster reserve. Ceiling: 60 nights × 24 × $22 ($31,680), 1,500 alarms × $30 ($45,000), two intake workers ($25,600), and $150,000 to the reserve. Charleston averages about 30 nights at or below 32°F from December to February; no published normal exists for 40°F, so 45 to 60 nights is an inference. Every unit price here is an estimate. The target fell by about three-quarters in verification, from $437,664 to $109,400, because the first draft priced heating help with a national cooling-bill average.

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Holy City Missions (Aldersgate warming shelter, 1444 Remount Rd, North Charleston)	Paid overnight staff, so the shelter opens every cold night (action 4)	$21,600	Direct grant	2 staff × 12 h × $20 × 45 nights; the wage is an estimate. The volunteer gap is documented (Live 5, 31 Dec 2025).	Partially confirmed
City of Charleston Fire Department and Berkeley County fire services	1,000 battery CO alarms with installation (action 5)	$25,000	Pre-positioned stock	~$25 list price per alarm (reseller prices). 88% of symptomatic CO poisonings were in homes without an alarm (CDC MMWR 2004).	Partially confirmed
Palmetto Community Action Partnership	Temporary intake staff, so more households reach LIHEAP before the freeze (action 3)	$12,800	Direct grant	1 FTE × 16 weeks; no staffing cost is published. Intake is the bottleneck: one 2022 release opened 78 Berkeley appointments.	Partially confirmed; cost unverified
Coastal Community Foundation, Lowcountry Disaster Response Fund	A Berkeley-designated reserve, released after a disaster	$50,000	Pooled fund	Matches the fund's largest recent grant ($50,000, Colleton). Precedent for a county designation: $150,000 for Dorchester from a 2024 Google gift.	Partially confirmed
Lowcountry Food Bank	Winter pre-positioning (action 8)	$0 until quoted	Pre-positioned stock	No usable unit cost. "$1 = 5 meals" is a fundraising ratio that relies on donated food, not the cost of bought stock.	Partially confirmed

The funder's one thing: the constraint here is people, not benefit dollars. Two overnight staff a night at Aldersgate, and caseworkers at Palmetto CAP, unlock more protection per dollar than another round of bill assistance. And very little of the money belongs on Daniel Island itself. Its residents are among the best placed in the region to ride out a storm; the money does the most good in North Charleston, on the peninsula, and in Goose Creek and Moncks Corner.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

Relying on "Turn Around Don't Drown" signs and messages to stop flood driving

CONTRAINDICATED PARTIALLY CONFIRMED

An LSU thesis (Bryant, 2021) found no convincing evidence that the campaign cut vehicle flood deaths in Texas since 1996. In a Tucson study, the most common reason drivers crossed was that the car ahead had made it. Here, the drowning on Omni Boulevard in December 2023 happened in a downpour, on a road the driver knew.

Instead: keep and extend Charleston's physical closures, staffed and enforced (action 1). Messaging works alongside a barricade, not in place of one.

Setting closures and alerts on the harbour tide forecast alone

CONTRAINDICATED PARTIALLY CONFIRMED

The October 2015 flood was rain-driven: 15 to 20 inches fell widely, about 18 on Daniel Island, while the harbour peaked at only 8.29 ft. On 17 Dec 2023 the tide came in about 1.6 ft above the final forecast. A tide-only trigger misses the first kind of flood and is late for the second. And a trigger set near 6 ft fires on ordinary spring tides, which teaches people to ignore it.

* Suggested formulation: check with a health worker before sharing.

Instead: trigger on a flash-flood warning during a high tide, keep the tide trigger at 7.5 ft, and plan for the tide to run about 1.5 ft above forecast.

Promoting generators without CO alarms*

CONTRAINDICATED CONFIRMED

In both CDC investigations cited in action 5, the deaths came from generators in the days after the storm, mostly overnight, in homes without a working alarm. A generator giveaway or "be ready with a generator" message without an alarm moves the risk from the cold to the bedroom.

Instead: pair every generator message with a battery CO alarm and "never indoors, in a garage, or near a window." Target the first three nights of an outage.

Opening warming centres and waiting for people to come

CONTRAINDICATED CONFIRMED

Transport is the barrier people name most often. 29492 has no warming site, intake closes at 9 pm, and on ice nights the Wando and I-526 crossings can close. Someone on the Cainhoy side cannot reach Aldersgate or Hibben without a ride.

Instead: extend the rides Charleston already runs (CARTA, Tri-County Link, the Citadel Square pickup) to the Wando side, and text the plan to employers by 6 pm so workers who commute in know the intake window.

Counting on the peninsula sea wall

CONTRAINDICATED CONFIRMED

The Battery extension, about $1.3 billion and shared 65/35 between federal and local money, has only design contracts ($6.22 million, signed from October 2025). Nothing is built. It protects the peninsula from surge, not from rain, and it does not cover Daniel Island or Cainhoy.

* Suggested formulation: check with a health worker before sharing.

Instead: spend this winter's effort on closures, drainage upkeep (the property owners' association already clears outfalls), and the people-centred actions above.

Spending the window asking SC DSS to write a disaster food plan

CONTRAINDICATED

Federal rules (7 CFR 272.2) already require the state to keep a current disaster SNAP plan and certify it each year. Lobbying for one wastes effort. The real gap is the switch: disaster SNAP runs only after a presidential major-disaster declaration with Individual Assistance, which winter storms seldom get.

Instead: ask SC DSS how food help would reach Berkeley and Charleston households after a nor'easter or ice storm that gets no such declaration, and back the food bank's winter stocking (action 8).

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Oct 2015 (El Niño building)	"1,000-year" rain flood: 15–20 in widely, over 25 in locally; about 18 in on Daniel Island; harbour 8.29 ft	17–19 deaths statewide (sources differ), many in vehicles; sewage overflows on Seven Farms Dr and nearby streets; Publix closed	New USGS flood maps; the 2019 Dutch Dialogues, which the city adopted but is still writing into zoning
2015–16 (strong El Niño)	Record run of tidal flood days at Charleston	No deaths recorded	NOAA tied the spike to El Niño on top of sea-level rise
Jan 2018 (La Niña)	Ice storm	Not recorded locally	Showed the access risk: I-526 ramps and the Wando bridge on Highway 41 closed

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YEAR	EVENT	TOLL	WHAT CHANGED AFTER
17 Dec 2023 (El Niño)	Nor'easter; harbour 9.86 ft, 4th highest since 1922 and the highest without a hurricane	1 death: Nancy Morrow, 72, drowned in her car on Omni Blvd, Mount Pleasant; 78 rescued, mostly from vehicles	Nothing structural found
Jan 2025	Over 4 in of snow	Not recorded locally	Airport and Ravenel Bridge closed
Jan 2026	Winter Storm Fern	1 death in Charleston County: a 59-year-old man, reportedly homeless, believed hypothermia; 6 statewide	No 2026–27 shelter plan published as of this briefing
4 Oct 2026	Downtown flash flood	Nine vehicles submerged; no deaths reported	Barricade and ticketing enforcement on downtown streets

The recurrence nobody has noticed: Daniel Island stays dry because its private developer built above the public code. Roads were set at least 7.5 ft, a foot above the city standard, and drainage was sized for a 25-year storm rather than a 10-year one. That margin was set for spring tides and hurricanes, but the 2023 El Niño nor'easter came within inches of Irma's tide with no hurricane at all. The new building along Clements Ferry Road is not held to the island's standard, and the public promises (the 2019 Dutch Dialogues, the sea wall, the Spring–Fishburne drainage tunnels) have all slipped.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Trigger closures on the tide forecast** versus **trigger them on rain meeting a high tide**. The emergency manager and historian built from the tide record; the skeptic showed that the deadliest local flood (2015) and the 2023 drowning were rain-driven. Verification supports using both triggers. A season of logged closures against NWS warnings would settle the thresholds.
- **Build volunteer capacity on Daniel Island** versus **send the money off the island**. The frontline responder sees the property owners' association and Beeple Studios as ready-made networks; the philanthropist and public health scientist say the deaths will be in North Charleston and on the peninsula. Both hold: the island's networks cost nothing, and they can be the source of volunteers and money for the places at higher risk.

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- **Stock food in pantries** versus **put cash in households**. The food lens favours pre-positioned stock; its verifier found that stock helps only while roads are cut, and that SNAP and cash do more after that. The open question is how long a winter closure lasts here, and no record was found.

WHAT NOBODY COVERED

BLIND SPOT

Two groups inside 29492 went unexamined. The first is the Spanish-speaking construction and hospitality workforce along Clements Ferry Road; the one source that might have described it would not load. The second is the older Black communities of the Cainhoy peninsula, such as Cainhoy village and Huger, as distinct from Phillips across the Wando. No lens found who they trust or how they hear warnings, and the frontline actions above would reach neither group well. A local partner who knows those communities should review actions 4 and 6 before they run.

HOW TO READ THIS

METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** Two requests from the same person, on the same day, were answered together: "Beeple Studios, Charleston, SC" and "Charleston, SC". The studio's address geocodes to the Cainhoy side of ZIP 29492, so the location was set as ZIP 29492, Daniel Island and Cainhoy, Charleston, with the wider city as context. Climate Impact Lab impact region, as the report records it: a Berkeley County, SC region, inferred and not resolved. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February

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2027. It covers heat only, and it publishes no figure for the United States. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Heat is not this place's winter hazard, so the historical record carries the hazard instead of a borrowed number.

- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. Food actions were checked for whether their data series are actually published, for timing, and for cash versus in-kind help. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Could not be consulted. The page renders with no entries and no searchable content without a browser session, and its download was not reached in this run. No action here is marked as having or lacking a precedent there.
- **Reader.** Added 25 October 2026, after the requester asked for steps at a "take care of yourself" level: power, water, food, and communications. The For you steps are taken from the verified actions and from a separate check of household guidance for 29492 on the same three tests (3 confirmed, 6 corrected). The rest of the report is written for the city, the counties, and their partners.
- **Health wording.** Passages where a wording error could hurt someone (flooded roads, carbon monoxide, generators, medical equipment in an outage, warming sites, emergency water, and vaccines) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing

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them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. The impact region was not resolved on the Impact Map. Post and Courier pages often returned HTTP 429, so several Charleston details rest on other outlets' reports. The NWS forecast for 17 Dec 2023 and the city's 7.0/7.5/8.0 ft tide tiers come from secondary sources and the city's TideEye page, not from an NWS document. The SC utility regulation text was not retrieved; the medical-certificate and cold-weather rules come from the Office of Regulatory Staff's consumer sheet. A Foodtown grocery reference turned up by one lens was found to be in Charleston, West Virginia, and was removed.

References · Flooding and tides: NWS Charleston, 17 Dec 2023 nor'easter (confirmed) · NWS Charleston, October 2015 flood (confirmed) · Mean High Water, Dec 2023 forecast vs. observed (secondary) · City of Charleston TideEye (confirmed) · Omni Blvd drowning, Gray syndication (confirmed) · Live 5, coroner and 78 rescues (press) · Live 5, 4 Oct 2026 flash flood (press) · NOAA, 2015–16 tidal flooding and El Niño (confirmed) · Daniel Island News, October 2015 (confirmed) · Daniel Island News, island drainage standards (secondary) · Daniel Island News, flood zones (confirmed) · Bryant 2021, LSU thesis on Turn Around Don't Drown (confirmed; not peer-reviewed) · CDC, Hurricane Matthew deaths (confirmed; North Carolina) · City of Charleston, Battery Extension · ABC News 4, $6.2M design contracts (confirmed)
Health and power: Lurie et al. 2015, dialysis after Sandy (confirmed) · HHS emPOWER public data, 7 Oct 2026 (confirmed) · emPOWER outreach dataset rules (confirmed) · IPRO ESRD Network 6 (confirmed) · Anderson and Bell 2012, 2003 blackout mortality (confirmed) · SC ORS, utility safety net (confirmed) · CDC MMWR 2004, CO after the 2002 ice storm (confirmed) · CDC MMWR 2005, CO after the 2004 hurricanes (confirmed) · CDC MMWR 2025, cooling-centre barriers (confirmed) · Charleston Fire, smoke alarm programme (confirmed)
Shelter and community: Charleston City Paper, Winter Storm Fern deaths (confirmed) · City of Charleston, January 2026 warming sites (confirmed) · Live 5, Salvation Army warming centre (confirmed) · Live 5, Aldersgate volunteer gap (confirmed) · Berkeley County warming centre (confirmed; January 2025) · Daniel Island News, POA storm plan (confirmed; 2019) · Berkeley County Emergency Notification System (confirmed) · Post and Courier, Beeple Studios 2026 events (confirmed) · SELC, Phillips Community Association (confirmed) · ABC News 4, 2018 road and bridge closings (press)
Money and food: CRS IN10890, NFIP take-up in the 2015 floods (confirmed) · FEMA bulletin W-13017, NFIP waiting period (confirmed) · SC OEO, LIHEAP providers (confirmed) · Live 5, LIHEAP appointments 2022 (confirmed) · Berkeley Electric Operation Round Up (confirmed) · SC parametric insurance RFI 5400030048 (confirmed; secondary listing) · Coastal Community Foundation, disaster fund (confirmed) · CCF, Google gift (confirmed) · Lowcountry Food Bank, disaster response (confirmed) · SC DSS, SNAP by county, July 2026 (confirmed) · 7 CFR 272.2, state disaster plans (confirmed) · USDA ERS food price outlook, via RFD-TV (secondary)
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

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Generated 22 October 2026 for ZIP 29492, Daniel Island and Cainhoy, Charleston. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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